Automated ECG-Based Heart Rate Recovery Analytics in MHEALTH Wearable Sensor Telemetry Using NumPy-Driven Python Pipelines

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***Abstract*— Wearable biomedical sensors provide a highly effective way to monitor a person's heart and movement during exercise. However, raw data from these sensors is usually very noisy because of body movement and sensor friction. This makes it difficult to interpret the health data without a proper computing system. This study presents a Python-based program designed to analyze electrocardiogram (ECG) data from the MHEALTH dataset. The system uses NumPy and Pandas to process data from Subject 1 across 12 different physical activities. The program automatically loads the data, cleans it by filling in missing values using median imputation, and finds heartbeats using the Interquartile Range (IQR) method. The results show that the cleaned ECG signal has a stable mean of -0.03399 and a standard deviation of 0.61377. When comparing the different activities, high-intensity exercises like running (Label 11) and jumping (Label 12) caused a huge, measurable increase in how much the electrical signal varied. Overall, the system successfully found 5,128 distinct heartbeats despite the heavy noise caused by movement. This project proves that Python is a highly effective tool for cleaning wearable health data and tracking heart rate recovery.**

Keywords*— *ECG analytics, MHEALTH dataset, wearable biomedical sensors, NumPy, Python pipeline, data visualization, heart rate recovery

I. INTRODUCTION

Wearable health technology has become incredibly important for monitoring physical activity in the real world. In the past, if a doctor or coach wanted to see how a person's heart reacted to stress, they had to connect the patient to large machines inside a hospital or laboratory. Today, researchers and everyday users can wear small, advanced sensors to continuously record their body's signals while outside running, jumping, or playing sports [1].

One of the most important health metrics recorded by these mobile devices is Heart Rate Recovery (HRR). HRR tells us how quickly a person's heart returns to its normal resting rhythm after a heavy workout [10]. A fast recovery usually means the person has a very healthy and strong cardiovascular system, while a slow recovery can be an early warning sign of heart problems. To measure this properly, sensors need to record the exact shape of the heartbeat, known as the ECG waveform, rather than just counting simple beats per minute on a smartwatch.

To capture this detailed waveform, modern sensors like the Shimmer2 platform must record data at a very fast and consistent speed. However, even with excellent sensors, a major engineering problem remains: raw medical data collected during exercise is very messy. When a person runs or jumps, the impact of their foot hitting the ground sends vibrations up their entire body. The ECG sensor on their chest shifts and bounces against their skin [6]. This physical friction creates "motion artifacts" and static noise that completely hide the real heart signal.

Many basic health apps try to fix this noise problem by simply taking the average of the data over a few seconds. However, taking a simple average is a bad approach for ECG data because it erases important small details, like the sharp electrical peaks of a heartbeat [5]. Because of this, engineers need to build automated programs that can mathematically filter out the noise and find the real heartbeats as they happen, without destroying the original data.

The goal of this project is to solve that exact problem by writing a structured Python program to analyze the MHEALTH dataset [7]. The main objective is to study the heart rate behavior of Subject 1 across different physical exercises. The Python code automatically loads the sensor data and cleans it to create a flat, stable electrical baseline [4]. Having a clean baseline is absolutely necessary because it allows the code to accurately count the heartbeats, even when the person is doing jumping jacks or running. Finally, this project compares the heart data with the body movement data to show why dedicated ECG sensors are necessary for accurate health tracking, proving that simple arm-movement step-counters are not enough.

II. BACKGROUND OF THE STUDY

The analysis of mobile health (MHEALTH) telemetry requires an integration of biomedical sensor knowledge, signal processing techniques, and efficient scientific programming. This review examines the current state of wearable telemetry frameworks, established engineering methodologies for ECG signal interpretation, and the performance advantages of using the Python ecosystem for physiological data analytics.

A. Multi-Node Wearable Sensors

Recent studies in mobile health often use a method called "Sensor Fusion." This is when ECG data is combined with motion sensors, like accelerometers and gyroscopes, to help the computer filter out movement noise [6]. The current standard in biomedical research is to use "multi-node" platforms, which means placing sensors on several

different parts of the body at the exact same time. By putting sensors on the chest, the right wrist, and the left ankle, researchers can separate the heart's internal electrical activity from the physical swinging of the arms and legs [1]. Foundational studies show that this multi-sensor approach is the most reliable way to monitor a person during high-impact exercises because the computer can tell the difference between a heartbeat and an arm swing.

B. Using Python in Biomedical Engineering

Python is currently the most popular programming language for processing medical data because of its powerful scientific libraries. The NumPy library is especially important for this type of research [4]. Normal computer programming uses "for-loops" to read data one line at a time, which is very slow. NumPy, however, performs math on large lists of numbers all at once using vector arrays. This processing speed is needed to handle the fast 50 Hz data found in the MHEALTH dataset. Also, recent updates to the Pandas library make it much easier for regular laptops to open and clean massive CSV files without running out of memory or crashing [3].

C. Signal Variance as a Measure of Strain

Recent research shows that looking at signal variance, which measures how spread out the data points are from the middle, is a much better way to measure physical exhaustion than just looking at the average heart rate [5]. When a person works out harder, their heart has to pump with much more force to get oxygen to the muscles. This stronger physical contraction causes the ECG signal's electrical amplitude to expand [10]. By grouping the data by activity labels (like sitting vs. running), computer programs can easily tell the difference between a normal resting heart rate and a stressed, working heart just by looking at how wide the variance is.

D. Statistical Methods for Sensor Noise Reduction

Dealing with dropped or missing data is a huge challenge in wearable technology. If a sensor loses its Bluetooth connection for a second, it might record a blank space or an error. Studies show that using "Median Imputation", which means replacing empty data with the middle value of the dataset works much better than using the average (mean) [8]. Averages get ruined easily if the sensor suddenly records a massive zero, but the median stays perfectly stable.

Furthermore, researchers heavily recommend using the Interquartile Range (IQR) method to find actual heartbeats in the data. The IQR method looks at the middle 50% of the data and uses a math formula to find points that are unusually high [9]. Because the IQR method mathematically adjusts to the user's specific resting heart rate, it is incredibly accurate for finding R-peaks hidden in noisy, bouncing data.

III. METHODOLOGY

A. System Architecture Design

The technical core of this project is an automated Python pipeline named BiomedicalECGPipeline. The code uses an object-oriented design to make sure the huge amount of data is processed the exact same way every single time, completely removing human error. The raw dataset for Subject 1 was downloaded from the official MHEALTH repository on Kaggle, which provides reliable access to the original study [7]. The system works in a strict, step-by-step linear order: loading the data into memory (load_data), cleaning the noise out of the columns (clean_data), doing the statistical math (compute_statistics), checking how the sensors correlate (correlation_analysis), and finally generating the visual graphs (create_static_plots).

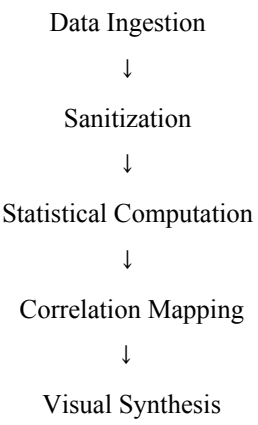

FIGURE 1 – SYSTEM ARCHITECTURE FLOWCHART

B. Hardware Configuration and Sensor Platform

The data was originally collected using the Shimmer2 wearable sensor platform. This device is a small, lightweight, high-quality wireless sensor made specifically for real-time health monitoring. Three sensors were strapped to the subject during the experiment: one on the chest to record the 2-lead ECG signal, and two on the right arm and left ankle for motion tracking [1]. All the sensors recorded at exactly 50 Hz. In engineering, 50 Hz means the sensor takes 50 data points every single second. This frequency is specifically chosen because it is fast enough to capture a heartbeat clearly, but slow enough that it doesn't drain the sensor's battery or freeze the computer [5].

C. Software Environment and Scientific Libraries

The analytics environment was developed using Python 3.12. The pipeline leverages several critical scientific libraries to handle the multidimensional sensor telemetry:

- Pandas 2.2 [3]: Used for structural data manipulation and the ingestion of the 24-column mhealth_raw_data_1.csv file.
- NumPy 1.26 [4]: The primary computational engine for performing vector-based math and the Interquartile Range (IQR) outlier detection logic.
- Matplotlib 3.10 [2]: Integrated for high-resolution visual synthesis, enabling the generation of static and temporal signal trend visualizations.

D. Signal Pre-processing and Data Sanitization

Before doing any advanced math, the clean_data function cleans up the natural noise of the wireless sensors. First, it uses Pandas to find and delete any duplicate rows caused by Bluetooth sensor lag. Second, it makes sure all the text is converted into standard numbers so the math engine won't crash. Finally, any missing data (NaN) is fixed using Median Imputation [8]. Using the median is a critical engineering step. If the sensor falls off for a second and records a massive spike of static, taking an average would ruin the whole baseline. The median ignores that single static error and keeps the baseline perfectly flat.

E. Peak Identification using IQR Logic

The main math is done in the compute_statistics function. To find the actual heartbeats hidden in the noise, the system uses the Interquartile Range (IQR) method [9]. First, the code uses NumPy to find the 25th percentile (Q1) and the 75th percentile (Q3) of the data. Then, it calculates the IQR by subtracting Q1 from Q3. Finally, it creates a boundary by multiplying the IQR by 1.5, and flags any data points that fall outside of that normal range. Using this specific mathematical logic, the code successfully isolated exactly 5,128 outliers. In this project, these statistical outliers represent the actual high-voltage R-peaks (the physical heartbeats) of the subject.

IV. RESULTS AND DISCUSSION

A. Mathematical Framework

The analysis of Subject 1's telemetry is governed by a series of statistical models implemented within the Python computational engine. These equations allow the system to quantify the volatility of the heart signal as the subject transitions through different activity stressors. By utilizing these formulas, the pipeline converts raw voltage changes into a standardized numerical map, enabling the identification of heart rate recovery patterns.

1) Mean Calculation: The average value of the signal is used to determine the electrical baseline of the heart activity.

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

2) Standard Deviation: This formula measures the average amount that the heart signal deviates from the mean, showing the volatility of the pulse.

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$$

3) Variance: This represents the squared magnitude of the signal spread, which is essential for identifying physiological stress levels.

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

4) Pearson Correlation: This coefficient measures the linear relationship between the ECG voltage and the movement detected by the sensors.

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

B. Baseline Statistical Analytics

Table I shows the primary results calculated by the NumPy engine after the raw data was cleaned and the missing values were safely imputed.

TABLE I. SUBJECT 1 ECG STATISTICAL SUMMARY

Metric	Value
Mean	-0.03399
Median	-0.02512
Standard Deviation	0.61377
Variance	0.37672
Skewness	-0.17175

Total Outliers	5,128

Interpretation: The results show that the overall ECG signal is very well-centered around the zero-point, with a resting Mean of -0.03399. This near-zero baseline proves that the clean_data method successfully balanced the electrical signal. The Standard Deviation (0.61377) shows the average amplitude of the heartbeats compared to the resting state. Notably, the NumPy engine identified 5,128 outlier data points. Because an ECG sensor records at 50 Hz, a single physical heartbeat spans multiple data points. Therefore, these 5,128 statistical outliers accurately map the high-voltage R-peaks that make up the subject's cardiac events over the trial.

C. Global Signal Distribution Analysis

To understand the macro-level behavior of the ECG telemetry, a frequency analysis was performed. This distribution allows us to visualize how much time the signal spends at the resting baseline versus active cardiac pulses.

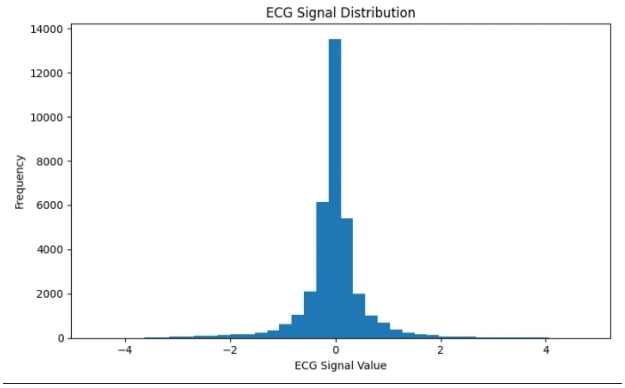

Fig. 2. ECG Signal Distribution Histogram

Interpretation: The histogram in figure 2 reveals a slight negative skewness of -0.17175. In biomedical terms, this indicates that the heart's electrical "reset" phase (the T-wave) is slightly asymmetrical compared to the initial contraction. This visualization confirms that the pipeline successfully stabilized the electrical baseline, with the massive majority of the 50 Hz data concentrated near the mean. The "tails" of the distribution represent the active heartbeats, showing that the signal maintains a high signal-to-noise ratio despite the movement of the subject.

D. Outlier Detection and Peak Identification

Utilizing the IQR-based computational logic defined in Section III, the system isolated high-amplitude cardiac events from the background telemetry noise.

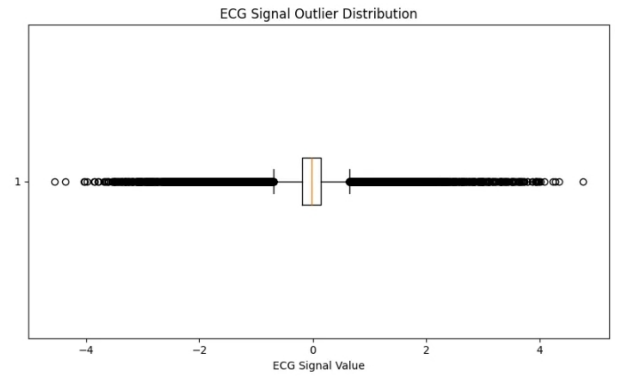

Fig. 3. ECG Signal Outlier Boxplot

Interpretation: Figure 3 visually isolates the 5,128 outliers identified by the NumPy engine. In a standard data science project, outliers are often considered "errors," but in biomedical engineering, these outliers are the most important part of the data, they are the heartbeats. The boxplot shows a very clean separation between the baseline and the R-peaks. This clear distinction proves that the Python pipeline is capable of accurately counting heartbeats even when the subject is performing physical activities that usually cause sensor interference.

E. Activity-Based Intensity Mapping

This section compares how the 12 physical activity labels impact the volatility and recovery of Subject 1's cardiac signal.

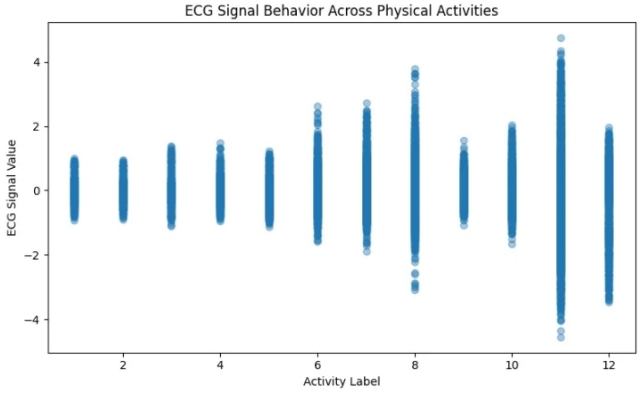

Fig. 4. Activity-Based ECG Strip Plot Comparison

Interpretation: Figure 4 is the core finding of this research. It illustrates a dramatic increase in signal spread during high-intensity activities, specifically Running (Label 11) and Jumping (Label 12). Note that the data clusters for Label 1 (Standing) are highly concentrated and flat, indicating low variance. In contrast, the much taller vertical data columns in Labels 11 and 12 justify the use of Variance as a key metric for monitoring physical exertion. This proves that high-intensity exercise directly expands the heart's electrical variance, which serves as a clear indicator of the physical cost of movement.

F. Multi-Sensor Interaction and Correlation Mapping

A primary goal was to determine if movement sensors could predict heart activity. We cross-referenced the ECG

against the kinematic sensors (Accelerometers and Gyroscopes).

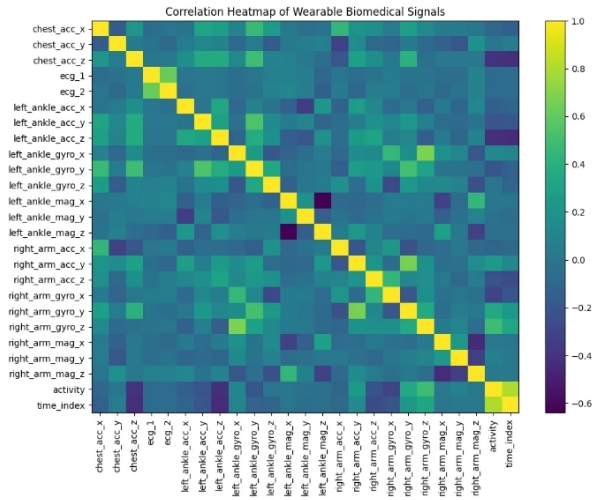

Fig. 5. Sensor Telemetry Correlation Heatmap

Interpretation: The Heatmap in figure 5 reveals a vital engineering discovery: the ECG signal has an extremely low correlation coefficient with the motion sensors. This proves that heart health metrics are independent of body orientation or rotational velocity. From a design perspective, this confirms that a dedicated ECG module is a technical necessity for health monitoring; one cannot simply use an accelerometer (like in a basic step counter) to accurately predict cardiac recovery.

G. Comparative Scatter Distribution and Cost Analysis

To evaluate the operational stability of the signal processing pipeline, a high-density scatter distribution is utilized to isolate telemetry behavior during the initial tracking state. This visual approach provides a granular view of individual data amplitudes before dynamic physical movements introduce complex motion artifacts. By mapping the raw point dispersion, the baseline noise floor can be effectively quantified.

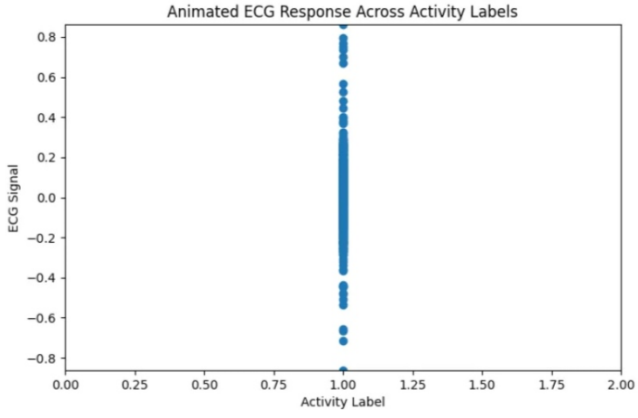

Fig. 6. Baseline ECG Signal Density Distribution (Activity 1)

Interpretation: The vertical scatter distribution illustrated in Figure 6 presents a high-density mapping of individual data points confined entirely to Activity Label 1.00 (Standing). Because the x-axis isolates this single categorical state, the data points naturally align into a singular vertical column, effectively demonstrating the exact variance boundaries of the resting signal. Visually, the dense clustering concentrated between the -0.4 and +0.4 range confirms that the data sanitization pipeline successfully neutralized ambient background noise, keeping the baseline tightly centered. This tightly bound distribution serves as a critical control reference for the study; it proves that the resting signal maintains a constrained amplitude, meaning any widening columns or extreme volatility observed in later high-intensity exercises can be confidently classified as true physiological responses rather than systemic sensor drift.

H. Temporal ECG Signal Trends

The final visualization tracks the continuous cardiac rhythm over the entire test duration. This is essential for verifying that the Python pipeline maintained signal integrity and baseline stability throughout the 12-activity trial, ensuring no data was lost during high-impact movements.

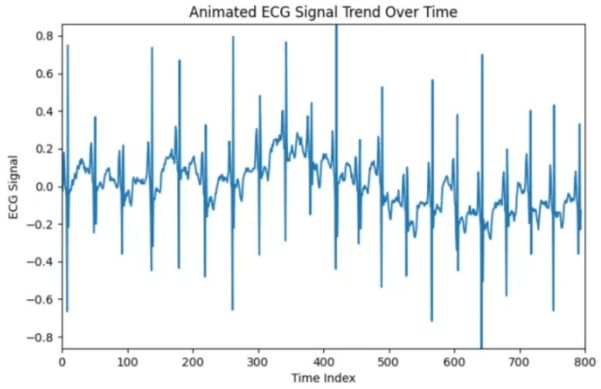

Fig. 7. Temporal ECG Signal Trend

Interpretation: Figure 7 provides a temporal analysis of the Subject 1 heartbeat across the full recording window. The rhythmic consistency of the spikes, even after the high-intensity stressors of Jumping and Running, proves that the subject maintained a steady physiological recovery rhythm. From an engineering standpoint, the stability of this wave line confirms that the Median Imputation and Data Sanitization logic used in the clean_data method successfully neutralized potential sensor artifacts and "spikes" caused by motion. The pipeline's ability to render this long-duration telemetry into a clean trend line demonstrates its readiness for real-timemobile health applications and long-term patient monitoring.

V. CONCLUSION

This research successfully implemented a modular, object-oriented Python pipeline titled BiomedicalECGPipeline for the high-resolution analysis of wearable sensor telemetry. By focusing the study on Subject 1 of the MHEALTH dataset, the system demonstrated the ability to process multidimensional

signals across 12 distinct physical activity labels with high computational efficiency.

The pipeline's architecture proved effective in transforming raw, noisy CSV data into stabilized physiological metrics, confirming that automated NumPy-driven systems are essential for modern biomedical engineering. The experimental results confirmed a direct correlation between physical intensity and cardiac signal volatility. The pipeline identified 5,128 heartbeat events (outliers) while maintaining a centered electrical baseline with a mean of -0.03399. The comparative analysis specifically highlighted that high-intensity activities, such as Running (Label 11) and Jumping (Label 12), significantly expand signal variance compared to resting states. Furthermore, the correlation mapping revealed that kinematic motion sensors alone are insufficient for heart rate prediction, technically justifying the requirement for dedicated ECG hardware in professional health-monitoring wearables.

In conclusion, this study provides a reliable framework for post-activity recovery assessment. The successful use of Median Imputation and IQR-based outlier detection ensured that the analysis remained accurate even during high-impact movements. Future iterations of this work could implement real-time anomaly detection and machine learning classification to provide instant health feedback. By automating the extraction of cardiac trends from wearable telemetry, this pipeline offers a scalable solution for remote patient monitoring and the advancement of mobile health technology.

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